DIGITAL HEALTH VALIDATION LAB

University of Glasgow

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Evaluating the environmental sustainability of AI in radiology: A systematic review of current practice

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Introduction

Data- and energy-heavy artificial intelligence (AI) technologies are increasingly applied in radiology, often without consideration of their potential environmental consequences.

We aimed to assess current practice in evaluating environmental sustainability (ES) impacts of AI-enabled clinical pathways in radiology.

Methods

- We systematically searched MEDLINE and Embase on 5th November 2024 for **clinical radiology studies using AI** to aid in radiological diagnosis or intervention **which discussed ES impacts**
- We included peer-reviewed, English-language studies from 2015 onwards
- Our **primary outcome** was any quantitative reporting of ES impacts (including carbon emissions, energy/water/mineral usage, waste generation/disposal, and impacts on material environments)
- Our **secondary outcome** was any within-text discussion of ES impacts
- We used Synthesis Without Meta-analysis (SWiM) based on effect size for our primary outcome, with our secondary outcome synthesized narratively

Study Protocol

Study characteristics

Figure 1: PRISMA Diagram

4,365 titles/abstracts screened → 4,230 records excluded
↓
135 full texts screened → 119 records excluded
↓
16 studies included → 6 studies reporting primary outcome

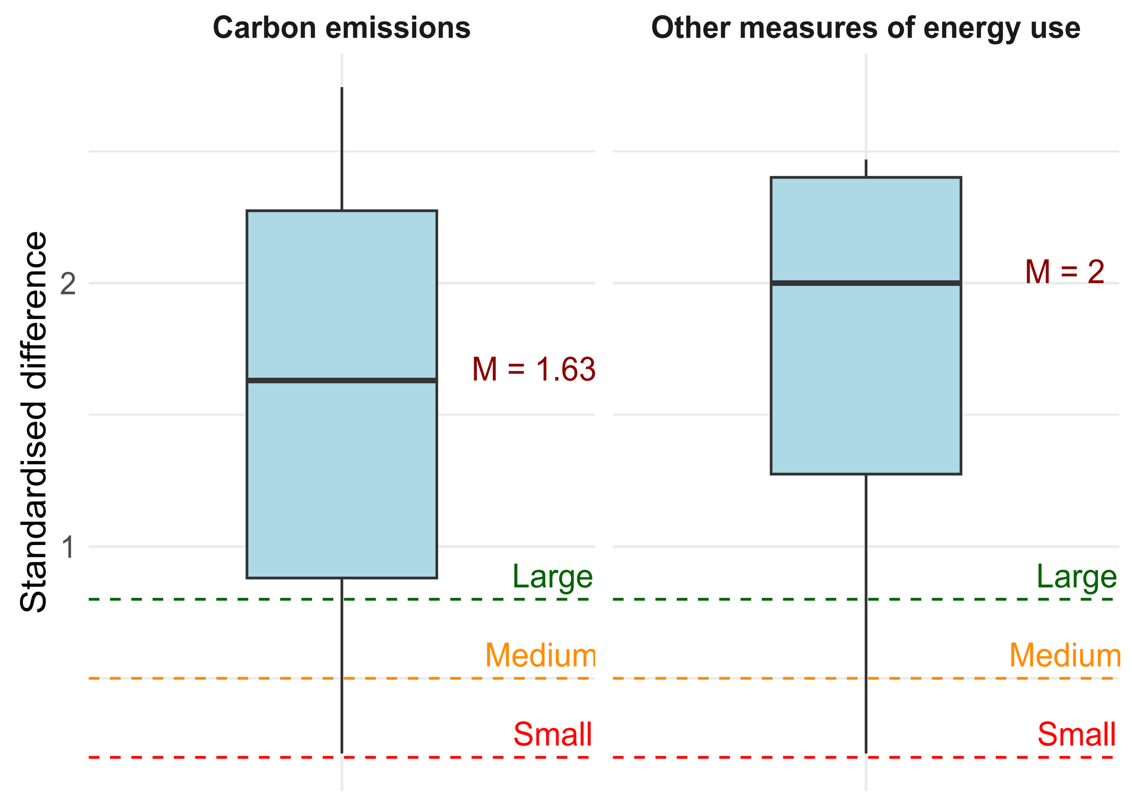
- Of 16 included studies, only 6 reported quantitative ES measures:
  - 3 reported on **carbon emissions**
  - 5 reported on **energy use**
- MRI most common modality (68.8%)
- **Most frequent AI tasks**: classification (43.8%) and segmentation (31.3%)
- **Most common architectures**: Green Learning (31.3%) and U-Net (37.5%)

Results

- All 6 quantitative studies evaluated algorithms designed to be resource/time efficient
- For all datapoints, **study algorithms outperformed state-of-the-art comparators on both environmental and technical performance**, producing
  - From **2.2 to 17.2 times less carbon emissions** (median 7.8)
  - From **1.6 to 751.6 times less energy consumption** (median 3.2)
- Standardised effect differences were almost all classed as **large** (Figure 2)
- No studies compared ES outcomes for AI vs standard-of-care
- 70% of studies for our secondary outcome included just a single sentence on sustainability

Results

Figure 2: Standardised differences of lightweight vs non-lightweight algorithms



Each plot represents 16 datapoints. M = median. Lines indicate Cohen's classification of standardised mean difference sizes.

Discussion

- Despite increasing concern about the climate impacts of AI, **environmental outcomes are rarely measured** or discussed within evaluations of AI-enabled clinical pathways in radiology
- However, our evidence suggests **designing AI products with sustainability in mind can substantially reduce their carbon footprint**, and highlights that more sustainable AI products already exist
- There remains a **large evidence gap** around comparing climate impacts of AI versus no AI in clinical pathways
- **Environmental sustainability should be better integrated into AI evaluation and procurement** to ensure costs and benefits for both climate and health are fully considered

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